

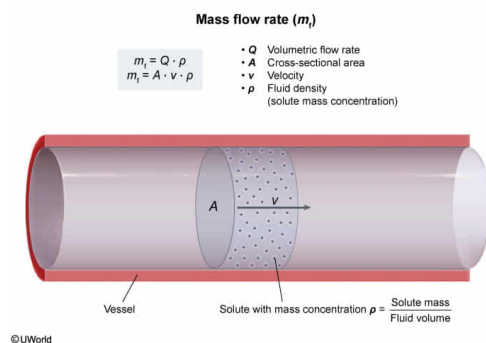
Blood flows through a vessel in the leg at an average velocity of 10 cm/s. How much glucose passes by a segment of the blood vessel each second if the radius of the blood vessel is 0.20 cm and each 1.0 cm³ of blood contains 1.0 mg of dissolved glucose?

- ☐ A. 4.0×10^{-1} mg/s [20%]
☐ B. 4.0×10^0 mg/s [14%]
☐ C. 1.3×10^{-1} mg/s [24%]
☒ D. 1.3×10^0 mg/s [40%]

Correct

41% answered correctly

Explanation:



Fluid flow describes the motion of liquid or gas molecules. For a fluid traveling through a closed conduit (eg, pipe), the volume of fluid flowing past a fixed point per unit time is the **volumetric flow rate (Q)**. Q is equal to the product of the velocity of the fluid (v) and the cross-sectional area (A) of the conduit at a given point:

$$Q = A \cdot v$$

Fluid flow may be further characterized by the movement of fluid mass per unit time. For a noncompressible, **ideal fluid** with constant density (ρ), the mass flow rate (m_f) quantifies the amount of mass flowing past a point within the conduit per unit time:

$$\frac{\text{Mass}}{\text{Time}} = \frac{\text{Volume}}{\text{Time}} \cdot \frac{\text{Mass}}{\text{Volume}}$$

$$m_f = Q \cdot \rho$$

The concept of mass flow rate may also be extended to solutes within a fluid, provided that solutes are **uniformly distributed** through the fluid.

In this question, blood with a glucose **mass concentration** of 1 mg/cm³ (ie, 1 cm³ of blood contains 1 mg of glucose) flows through a blood vessel with a radius (r) of 0.20 cm. Accordingly, the cross-sectional area of the artery is:

$$A = \pi \cdot r^2$$

$$A = \pi \cdot (0.20 \text{ cm})^2 = (3.14...) \cdot (0.040 \text{ cm}^2) = 0.126 \text{ cm}^2$$

Given a blood velocity of 10 cm/s, Q is:

$$Q = 0.126 \text{ cm}^2 \cdot 10 \frac{\text{cm}}{\text{s}} = 1.26 \frac{\text{cm}^3}{\text{s}}$$

Furthermore, the mass flow rate of glucose (m_{glucose}) at a segment of the artery is equal to the product of the volumetric flow rate and the blood glucose mass concentration:

$$m_{\text{glucose}} = 1.26 \frac{\text{cm}^3}{\text{s}} \cdot 1 \frac{\text{mg}}{\text{cm}^3}$$
$$m_{\text{glucose}} = 1.26 \frac{\text{mg}}{\text{s}} \approx 1.3 \frac{\text{mg}}{\text{s}}$$

Therefore, 1.3 mg of glucose passes by a segment of the artery each second (**Choice D**).

Educational objective:

A mass flow rate refers to the quantity of fluid or solute mass that flows past a fixed point per unit time. For solutes uniformly dissolved within fluids, the mass flow rate of the solute is equal to the product of volumetric flow rate and mass concentration of the solute (ie, mass of solute per unit volume of fluid).

Time spent:0 sec

Subject:

Physics

QID:402624

System:

Fluids